



Genetic improvement of farmed tilapias: Response to five generations of selection for increased body weight at harvest in *Oreochromis niloticus* and the further impact of the project

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ABSTRACT

Selection for increased body weight at harvest in Nile tilapia (*Oreochromis niloticus*) was carried out for five generations from 1991 to 1996, as a part of the project Genetic Improvement of Farmed Tilapias (GIFT). The base population for selection was composed of a mixture of various three- and four-way cross individuals descending from four wild African strains and four farmed Asian strains. Methods for single pair mating, separate rearing of full-sib groups and individual tagging and pedigree recording were developed. The parents for a new generation were selected based on a selection index including their own age corrected-body weight at harvest as well as that of their full- and half-sibs. Across generations, a total of 512 males were mated in a nested mating design to 941 females to produce 81,429 tagged progenies. Of those, 56,633 progeny had their body weight recorded at harvest after grow-out testing in a diverse range of farming environments. Estimates of the within generation realized response to selection across farming environments were obtained as the difference between the least squares mean performance of offspring of selected parents and of offspring of parents with average values of the selection index. The average realized response to selection per generation was 13.6% (range 9.0 to 20.1%), resulting in an accumulated response over five generations of 88% relative to the base population. A genetic trend analysis based on BLUP breeding values estimated across generations after the termination of the project suggested an accumulated response of 67%. The genetic composition of the synthetic population also changed during selection. The proportion of ancestors from three of the wild African founder strains increased from 59.7 to 76.3%. The accumulated coefficient of inbreeding was 7.1%. After the termination of the GIFT project, selection was continued in the GIFT population. The population has also been used as a genetic source in a number of similar public and private selection programs. A status review of the presently recorded dissemination of the genetic material and methodology is presented.

Statement of relevance: The GIFT project showed that five generations of repeated selection based on testing of individually tagged and pedigreed individuals from a synthetic farmed population of Nile tilapia gradually increased the body weight at harvest. The total increase was 67–88%. Reports from several descending populations shows that the selection response has continued during >10–15 additional generations.

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1. Introduction

An international collaborative project to improve the genetic performance of farmed Nile tilapia (*Oreochromis niloticus*) was initiated at the Bureau of Fisheries and Aquatic Resources/Freshwater Aquaculture Center (BFAR/FAC) facilities in Muñoz, Philippines in 1988 (Gjedrem, 2012; Pullin et al., 1991). The project, commonly known by the name Genetic Improvement of Farmed tilapias (GIFT), lasted for 10 years until 1997 and was based on the experiences from the pioneering Norwegian salmon selection program initiated in 1971 (reviewed by Gjedrem, 2010). The initial objective of the project was to apply up to date farm animal breeding and selection technology to improve the performance of tropical farmed finfish. Nile tilapia was chosen as a test case because of the short generation interval of the species, and for its significance for a wide range of fish farming operations, including small-scale and backyard farmers. Growth performance was chosen as the selection goal because of its economic importance for the target farmers and because it may easily be recorded in the live test fish. The GIFT project was started by collecting genetic material from eight different sources, four wild populations in Africa and four farmed populations in Asia, that were reproduced in Muñoz and tested for growth performance as pure strains in eleven different farming environments (Eknath et al., 1993). A complete diallel cross experiment with all eight strains was then carried out to study possible heterosis effects on growth performance in seven different farming environments (Bentsen et al., 1998). It was concluded that genotype by environment interactions and non-additive gene effects were of minor importance for growth performance. Consequently, it was decided to form a synthetic population composed of three and four way crossed individuals involving all eight founder strains, and genetic parameters for growth performance in seven different farming environments were estimated (Eknath et al., 2007). The estimates suggested a low to moderate heritability for growth performance and high genetic correlations between growth performances in different farming environments (possibly somewhat lower correlations for one intensive cage farming environment). The parameter estimates were later confirmed by an analysis across five generations descending from the synthetic base population (Bentsen et al., 2012). Since the possible genetic gain from developing multiple environment specific strains or specialized hybrid parent strains seemed to be marginal, it was decided to focus on selection for improved additive genetic growth performance within the genetically diverse synthetic population, based on testing in mainly semi-intensive farming environments. Five generations of selection for improved growth performance were then carried out.

The GIFT project resulted in a faster-growing, synthetic aquaculture stock of Nile tilapia (Acosta and Gupta, 2010; ADB, 2004; Dey et al., 2000; Yosef, 2009) that has later been an important genetic source for a variety of public and private genetic improvement and/or dissemination programs (Eknath and Hulata, 2009; Ponzoni et al., 2010a). Although continuous genetic response to selection for growth performance has been reported in a number of descending populations (Gjedrem et al., 2012; Hussain et al., 2011; Luan, 2010; Ponzoni et al., 2005; Thodesen et al., 2011), the initial selection response during the GIFT project has only been preliminary reported (Bentsen et al., 2003; Eknath and Acosta, 1998; Eknath et al., 1998), and not yet been fully and scientifically documented. At the time of the GIFT project, the world aquaculture production of Nile tilapia was about 0.5 million metric tons and the focus was on small scale and backyard farmers. Since then, production has increased to 3.7 million tons in 2014 (as compared to 2.3 million tons of Atlantic salmon, FAO, 2016), and much of the production has evolved into large scale, commercial operations. The GIFT strain has been widely available and disseminated since the first generations of selection, and a significant proportion of the present production of Nile tilapia is likely to be based on genetic materials with some degree of GIFT ancestry (see Section 4.4 below). In this paper the procedures used and the results obtained during the first five

generations of selection for increased body weight at harvest in the GIFT project are documented, and the impact of the GIFT strain and technology is reviewed.

2. Materials and methods

2.1. The GIFT population

The initial generation of the GIFT population subjected to selection (G0) was composed of full- and half-sib progeny groups from a nested mating design (each male mated to two or more females). It involved 50 male breeders and 123 female breeders chosen from the best performing strain combinations (Eknath et al., 2007), out of the 64 combinations from the 8×8 complete diallel cross study (Bentsen et al., 1998). The genetic material used in the diallel cross study represented four wild African strains (Egypt, Ghana, Kenya, and Senegal) and four farmed Asian strains (Israel, Singapore, Taiwan, and Thailand) (Eknath et al., 1993). The parents of the G0 population were systematically mated to produce progeny that were various three- or four-way strain crosses between the eight strains. The genetic ancestry proportions from the founder strains to the G0 population varied from 3 to 27% depending on the growth performance of the strains, but under the restriction that all eight strains should be represented among the ancestors (confer Table 6). In subsequent generations, breeders were selected and mated according to a nested mating design without considering the pure strain ancestry of the breeding candidates, but matings between full and half-sibs were avoided. The number, age and harvest body weight of male and female parents used to produce the G0 population families and the families of the five following generations are given in Table 1. The performance and genetic characteristics of the base population (G0) were reported by Eknath et al., 2007 and of the following five generations (G1 – G5) by Bentsen et al., 2012.

2.2. Mating and rearing of fry and fingerlings

A detailed description of the methods applied for single pair mating and tagging of progeny was given by Eknath et al. (2007). In summary, the selected breeders of each sex were conditioned in separate net enclosures (called 'hapas') within a pond at a stocking density of 2 to 5 fish per m^2 for 1 to 2 days. They were then transferred to 1 m^3 hapas at a stocking density of 3 to 5 males or 5 to 7 females per hapa for two weeks. During this period, breeders were fed ad libitum twice daily with a mixture of 70% rice bran and 30% fish meal. Once the breeding pond was prepared and fertilized, each female breeder was transferred to a separate 1 m^3 breeding hapa installed within the pond. The spawning condition of all female breeders was evaluated at stocking by examining the female genital papilla (Longalong and Eknath, 1995; Longalong et al., 1999). Females with the most swollen papillas were given priority for mating by stocking a male of similar body weight into the breeding hapa, avoiding mating of full or half-sibs. After 10 to 14 days, fry were collected from each breeding hapa and stocked in separate 1 m^3 fine mesh nursery hapas at a density of 150 to 200 fry/ m^3 . The male was transferred to another breeding hapa to be mated with a new female. After 21 days of rearing, the fingerlings were transferred to 1 m^3 hapas with larger mesh size (B-net hapas) at a stocking density of 100 to 150 fish per hapa. Supplementary feeding was provided in the fry and fingerling hapas. All full-sib families were reared separately until tagging at a live weight of 3 to 5 g. Following a rearing period of 6 to 13 weeks, all the fingerlings were tagged within a three-week period, using modified Floy® external fingerling tags as described by Longalong et al. (1999). Equal numbers of tagged fingerlings from all full-sib families were pooled together and conditioned in a separate tank for each test environment for 1 to 2 days without feeding before communal stocking in the different test environments.

Table 1

Number, age and harvest body weight of selected males and females that contributed progeny in the different generations, and the period when swim-up fry were collected from the breeding hapas.

Generation	Male breeders				Female breeders				Fry collection period
	No.	Age (days)	Weight (g)		No.	Age (days)	Weight (g)		
			Mean	SD			Mean	SD	
G0	51	369	88	24	124	369	52	13	12 Jan. – 6 Mar. 1991
G1	99	234	121	17	181	235	89	21	11 Sep. – 21 Oct. 1991
G2	100	325	139	21	185	328	114	20	15 Jun. – 20 Aug. 1992
G3	91	378	185	33	162	377	116	26	30 Jun. – 31 Aug. 1993
G4	90	371	263	57	155	369	178	45	2 May – 10 Aug. 1994
G5	81	446	270	45	134	447	154	24	11 Sep. – 17 Nov. 1995

2.3. Testing for growth performance

Across generations, a total of 81,429 tagged fingerlings were stocked in two to eight different farm environments per generation. The number of test environments and number of replicates within each environment in each generation are given in Table 2. Characteristics of the test environments and the dates of stocking were presented in detail in Eknath et al. (1993, 2007) and Bentsen et al. (2012). The test environments included the following: standard fertilized earthen ponds at BFAR Satellite Stations in southern (S1) and north-western (S2) Luzon, P1 and P2 were standard fertilized ponds (P2 with supplementary feeding), W2 was a pond fertilized with ipil-ipil (*Leucaena* sp.) leaves, C2 was cage culture in a reservoir with supplementary feeding, and RF was a rice-fish culture system with trench refuges, all located by the BFAR and FAC research facilities in Muñoz. Finally, C4 was a cage culture fed with standard pellet feed at a commercial cage farm.

After a grow-out period of 89 days on average across test environments in G0 and G1, and 123–126 days on average in the later generations (Table 3), the test fish in all environments were harvested. Ponds were harvested by seining in a maximum of three drags followed by complete draining of the ponds. The number of individuals without tags was easily identified by scars below the dorsal fins. Across test environments, mortality rate was lowest (7.8%) in G3 and highest (30.8%) in G5 (Table 2). The proportion of tag losses was lowest (4.7%) in G1 and highest (23.2%) in G5. At harvest, sex ratio (% males) was lower than 50%, except for in G0 and G5. Mortality, tag loss and sex ratio within test environments in each generation are given by Eknath et al. (2007) and Bentsen et al. (2012). The harvested fish were held in aerated tanks with continuous flushing of freshwater for one to two days during recording and before the breeding candidates were restocked in ponds at the Muñoz research facilities. Harvest body weight and sex were recorded for all recovered and tagged test fish, in total, 56,633 individuals across generations. The number of fish recorded in each generation and the mean body weight and age at stocking and at harvest are presented in Table 3. The G0 and G1 test fish had lower mean body weight at harvest than the later generations, reflecting the shorter grow-out testing period. Information about the number of fish, body weights and age

within each test environment in each generation is given by Eknath et al. (2007) and Bentsen et al. (2012).

2.4. Selection of broodstock

Because of limited computing facilities at the project site at the time of the experiments, it was decided to base the selection of brood stock on records of harvest body weight, sex and sib relationship from contemporary individuals within each generation only. The parents of generations G1 to G5 were selected among the animals reared in the two test environments P1 and P2, and based on their predicted breeding values for harvest body weight obtained from a classical selection index including their own body weights as well as the body weights of their full and half-sibs in all test environments. The predicted breeding values for each generation were obtained as follows. First, the observed harvest body weights within the different levels of the combined environment and sex sub-groups (ExS, with 14 levels in G0, 12 in G1, four in G2 and G3, six in G4 and eight in G5, see Table 2) were pre-corrected within each generation to the same overall mean and standard deviation. This was done by multiplying the deviation of each observed body weight from the mean value of the ExS sub-group by the ratio between the mean of the standard deviations of the body weights in all levels of ExS and the standard deviation of the actual ExS sub-group and then adding the overall mean body weight across all levels of ExS (Hill, 1984). Second, these pre-corrected body weights were adjusted for the animals' age at harvest within each level of ExS according to the linear regressions of body weight on age at harvest in the sub-group.

The estimated breeding values (EBV) were then calculated using independent linear functions of the pre-corrected body weights (PBW) (Gjerde, 2005; Wray and Hill, 1989):

$$EBV_i = \mu_p + b_i(P_i - \bar{P}_f) + b_f(\bar{P}_f - \bar{P}_h) + b_h(\bar{P}_h - \mu_p)$$

where P_i is PBW of the breeding candidate, $\bar{P}_f$ is the mean PBW of its full-sibs including the breeding candidate; $\bar{P}_h$ is the mean PBW of its half-sib groups including the PBW of its full-sibs (i.e. $\bar{P}_h = (\bar{P}_f + \bar{P}_{h_1} + \bar{P}_{h_2} + \dots + \bar{P}_{h_{m-1}})/m$ and μ_p is the overall mean PBW. The selection criterion coefficient for each of the three parts in the above equation was calculated as:

$$b_i = \frac{\sigma_w^2}{\sigma_w^2 + \sigma_e^2}$$

$$b_f = \frac{\sigma_d^2 + \sigma_w^2/n_f}{\sigma_d^2 + \sigma_c^2 + (\sigma_w^2 + \sigma_e^2)/n_f}$$

$$b_h = \frac{\sigma_s^2 + \sigma_d^2/m + \sigma_w^2/(n_f + n_1 + \dots + n_{m-1})}{\sigma_s^2 + (\sigma_d^2 + \sigma_c^2)/m + (\sigma_w^2 + \sigma_e^2)/(n_f + n_1 + \dots + n_{m-1})}$$

where $\sigma_w^2 = 1/2\sigma_A^2$ and σ_A^2 is the additive genetic variance; $\sigma_s^2 = \sigma_d^2 = 1/4\sigma_A^2$ is the sire and dam additive genetic variance component, respectively; σ_c^2 is the variance component for the effect common to full-sibs

Table 2

Test environments used in each generation.

Generation	Test environments ^a (No. of replicates)							
	S1	S2	P1	P2	W2	C2	C4	RF
G0	1	1	2	2	1	1	1	
G1	1	1	2	2	1	1		
G2			2	2				
G3			2	2				
G4			2	2			1	
G5			2	2			1	1

^a S1 and S2: Ponds at BFAR Satellite Stations, P1, P2 and W2: Ponds at Muñoz research facilities, C2: Cages at Muñoz research facilities, C4: Commercial cage culture, RF: Rice-fish culture at Muñoz research facilities.

Table 3

Number of fish, body weight and age at stocking and harvest in each generation. Mortality, tag loss and sex ratio also shown.

Generation	No of tagged fish at		Lost records (%)		Sex ratio (% males)	Body weight (g) at				Age (days) at			
	Stocking	Harvest	Mortality	Tag loss		Stocking		Harvest		Stocking		Harvest	
						mean	SD	mean	SD	mean	SD	mean	SD
G0	19,467	13,567	16.8	13.5	50.9	4.6	1.9	60.1	25.0	78.4	13.9	167.5	14.9
G1	20,740	15,385	21.1	4.7	42.1	3.5	1.3	47.5	29.0	84.0	11.7	173.0	11.4
G2	10,171	8661	8.5	6.3	44.5	5.1	1.8	97.2	38.3	112.3	15.1	237.0	15.2
G3	9574	6808	7.8	21.1	43.2	3.8	2.2	146.2	63.4	84.5	15.1	207.7	15.4
G4	8822	6385	17.9	9.7	39.7	4.6	2.6	119.7	46.9	90.1	28.4	216.4	28.1
G5	12,655	5827	30.8	23.2	53.6	4.0	2.1	156.8	59.0	109.1	18.1	233.7	16.7

other than additive genetics, and $\sigma_e^2 = \sigma_P^2 - \sigma_A^2 - \sigma_c^2$ is the random environmental variance in which σ_P^2 is the total phenotypic variance; m is the number of dams mated to the actual sire; n_f is the number of recorded full-sibs and $n_1, n_2, \dots, n_{m-1}$ is the number of recorded half-sibs for each of the other dam mated to the actual sire.

In the above prediction equation of independent terms, the regression coefficients of each term can be obtained without the need to invert the phenotypic variance-covariance matrix of the observed body weights (i.e., the left hand side of the classical normal index equation) for each set of combinations of the number of dams mated to each sire and the number of sibs in each full-sib family.

For each generation an estimate of σ_A^2 and σ_c^2 for the pre-corrected harvested body weights were obtained from a sire and dam model, and from which σ_c^2 was obtained as the difference between the dam and the sire components of variance. The applied estimates of σ_A^2 and σ_c^2 ; i.e. of the heritability (h^2) and of the effect common to full-sibs ($c^2 =$

σ_c^2 / σ_P^2) varied over the generations, h^2 typically from 0.2–0.3 and c^2 from 0.05 to 0.10 depending on the estimates obtained from each of the generations.

The maximum number of sires and dams selected from the same full-sib family was restricted to 15. An excess number of top ranked breeders were selected and stocked for conditioning as described earlier, in order to achieve the target of 100 males hierarchically mated to 200 females in each generation. However, some of the pairs stocked for breeding did not succeed in producing swim-up fry in the breeding hapas. Because of the limited duration of the breeding and fry collection period, the number of parents contributing to the next generation was lower than the target, and more so in the later generations (Table 1). Mating between full- or half-sibs was avoided.

2.5. Control groups

During the selection experiment, body weight at harvest was affected by the test environments, the age at harvest and the stocking season, which varied considerably from one generation to another (Tables 1, 2 and 3). The response to selection could not be estimated from the mean phenotypic performance of the selection line in each generation. Consequently, a number of additional individuals representing the average genetic performance in each generation were also selected as breeders to be mated and produce progeny that served as internal control groups along with the progeny of the contemporary selection line breeders (Table 4). These breeders were single pair mated (each male mated to one female, and each female to one male) to produce a set of control group families that were tagged and communally stocked together with the selection line test fish in all test environments in the following generation. The response to selection could then be estimated as the difference in performance between contemporary selection line and control group families. The G1 control group was produced by re-mating a random sample of the parents of the G0 generation. The G2, G3 and G5 control groups were progeny of individuals with average

breeding values in the previous generation. The G4 control group was progeny of re-mating of surviving and reproductive parents of the G3 control group (thus reflecting the total response across G3 and G4). On average 22 control full-sib families (ranging from 12 to 43) with, on average, 33 sibs per family (ranging from five to 96) were tagged and stocked in each test environment along with the selection line families in each generation. There were equal numbers from each control family in each test environment. Note that the control groups were produced from each new generation of the selection line, and not established as a separate control line. However, a second repeated mating of surviving and reproductively able parents of the G1–G3 control families was carried out one generation later to produce control families that were tested along with the grand-progeny of contemporary breeders in the selection line, in order to improve the genetic ties across generations and thus obtain a more reliable prediction of the genetic gain in a genetic trend analysis across all generations.

2.6. Statistical analysis

The body weights at harvest were affected by a number of systematic effects that were not related to selection response. The number and choice of different test environments varied from generation to generation (Table 2) and the growth performance differed considerably between test environments (Bentsen et al., 2012; Eknath et al., 2007), e.g. from 18 g (environment C2) to 88 g (environment P2) in G1 (average of males and females, Bentsen et al., 2012). The duration of the grow-out test period also increased by >30 days from G0 and G1 to later generations. The average age at stocking and at harvest differed between generations (Table 3) depending on the time needed to successfully reproduce a sufficient number of breeding pairs (Table 1) and the duration of the grow-out test period. The association between age and body weight at harvest in the present material has been reported earlier (Bentsen et al., 2012; Eknath et al., 2007). It was concluded that this association was non-linear in many test environment by sex sub-groups. The average phenotypic body weight also differed between sexes (males 20% to 70% heavier than females depending on test environments, Bentsen et al., 2012; Eknath et al., 2007), and the sex ratio differed between generations (Table 3). Finally, in G0 and G5, the test period was mainly during the warmer part of the dry season (March to May, mean air temperature 30–35 °C), whereas G1 to G4 were mainly tested in the cooler part of the dry season (November to February, mean air temperature 25–30 °C). Consequently, the realized genetic gain was estimated by comparing age corrected least squares means of body weight at harvest in progeny of selected parents with internal control groups that were progeny of average parents in each generation. The genetic gain was also estimated from BLUP breeding values calculated across all generations, as described below.

2.6.1. Realized genetic gain in phenotypic mean values

Age corrected least squares means across test environment by sex sub-groups within generations for harvest body weight of the

Table 4

Origin of control groups used in the different generations, the number of control fish at stocking and harvest, and their body weight and age at harvest.

Generation	Parents of control group			Control group					
	Type of parents	No. of sires	No. of dams	No. of tagged fish present at		Body weight at harvest (g)		Age (days) at harvest	
				Stocking	Harvest	Mean	SD	Mean	SD
G1	Random parents of G0 ^a	11	12	757	484	38.4	24.0	81.7	11.7
G2	Average G1 parents	23	23	200	164	84.0	34.6	113.8	19.3
G3	Average G2 parents	44	43	197	139	132.8	56.4	84.0	15.8
G4	Average G2 parents ^a	17	16	975	756	121.1	48.0	121.9	22.8
G5	Average G4 parents	16	16	1541	782	131.7	47.4	109.1	15.0

^a Repeated spawning.

selection line and the control groups were obtained from the following linear model:

$$y_{ijk} = a + (ExS)_i + G_j + \beta_{1i} * age_{ijk} + \beta_{2i} * age_{ijk}^2 + e_{ijk}$$

where y_{ijk} is the observed harvest body weight, a is an overall constant, $(ExS)_i$ is the effect of test environment by sex sub-groups, G_j is the effect of genetic groups (selected or control) and β_{1i} and β_{2i} are the linear and quadratic regression coefficients within the test environment by sex sub-groups of age at stocking on body weight at harvest and e_{ijk} is the residual random error. The analysis was carried out across test environments, since the genetic correlations between body weights at harvest in different test environments were close to unity for all environments except C4 (Bentsen et al., 2012) (C4 was only included in the base population and the two last generations).

The genetic gain in percent per generation was then calculated by dividing the difference between the least squares mean of the selected (LSM_s) and control (LSM_c) groups by LSM_c . The genetic gain per generation measured in additive genetic standard deviations (σ_A) was also calculated as follows:

$$\Delta G(\sigma_A) = \frac{\Delta G}{\sigma_A} = \frac{LSM_s - LSM_c}{LSM_c * CV_p * h} 100$$

where the average phenotypic coefficient of variation (CV_p) and the square root of the heritability (h), both across generation by test environment by sex sub-groups, were obtained from Bentsen et al. (2012).

2.6.2. Realized genetic gain in BLUP breeding values

After the termination of the GIFT project, individual breeding values for body weight at harvest were re-estimated using best linear unbiased prediction (BLUP), fitting an uni-variate mixed model with the DMU software (Madsen and Jensen, 2008) for harvest body weights that were pre-corrected to the same standard deviation across all levels of the combined generation by test environment by sex effects (Hill, 1984). The model included the fixed effect of the combined generation by test environment by sex sub-classes and the first and second degree polynomials of age at harvest as covariates nested within the sub-classes, a random animal additive genetic effect, a random effect common to full-sibs other than additive genetics and a random residual effect, as described in more detail by Bentsen et al. (2012). Included in this analysis were all tagged individuals with records of harvest body weight from all the families in G0, all the selected families in the following five generations, and all the control families. The overall estimates of heritability ($h^2 = 0.16$) and other effects common to full-sibs ($c^2 = 0.10$) from Bentsen et al. (2012) were assumed. Mean breeding values for body weight at harvest in each generation of the selection line and in the control groups (separately for repeated matings) were calculated, to obtain a genetic trend estimate across all generations.

2.6.3. Strain ancestry proportions

Population G0 was composed of a mixture of different three- and four-way strain crosses of the eight original pure strains (four wild-caught African strains from Egypt, Ghana, Kenya and Senegal and four farmed Asian strains from Israel, Singapore, Taiwan and Thailand, Eknath et al., 1993, 2007). The pure strain ancestors of each individual parent used in each generation of the experiment were traced through their pedigree records and the ancestry proportion from each strain was computed for each generation.

2.6.4. Accumulated inbreeding

Coefficients of inbreeding were obtained for all individuals in all generations using DMU (Madsen and Jensen, 2008) from which the average coefficient of inbreeding in each generation was calculated. The coefficient of inbreeding was set equal to zero for the parents of the G0 individuals, since only their strains of origin, but not their individual parents, were known (Bentsen et al., 1998).

3. Results

The average phenotypic body weight at harvest in each generation increased throughout the selection experiment from 60.1 g in G0 to 156.8 g in G5 (Table 3). However, the body weights at harvest were affected by a number of systematic effects that were not related to selection response, as pointed out earlier. The number of tagged selection line individuals that had their body weight recorded at harvest in each generation varied from 5827 to 15,385 depending on the number of test environments used in each generation and the mortality and tag loss (Table 3). The number of selection line parents that contributed progeny from single pair matings in a nested mating design in each generation was on average 85 males (range 51 to 100) and 157 females (range 124 to 185) (Table 1).

3.1. Realized genetic gain in phenotypic mean values

The effects of test environments, sex and age on body weight at harvest within generations were accounted for according to the previously described linear model in Section 2.6.1. The least squares means for body weight at harvest were significantly heavier ($P < 0.01$) in progeny of selected parents in all generations, than in the control groups of progeny of parents with average genetic performance for body weight at harvest (Table 5). The average realized response to selection per generation was 13.6%, but variable across generations (range 9.0 to 20.1%). The average realized response per generation measured in additive genetic standard deviations was 0.99. The accumulated realized response over five generations of selection was 88% relative to the average performance in G0 (Fig. 1).

Table 5

Least squares means of the selection line and control groups and genetic gain in the selection line relative to the control groups measured as a % and in additive genetic standard deviations ($\Delta G(\sigma_A)$).

Generation	LSM $\pm$ s.e of body weight at harvest (g)		Genetic gain per generation	
	Control	Selected	%	$\Delta G(\sigma_A)$
G1	41.8 $\pm$ 0.8	50.2 $\pm$ 0.2	20.1	1.46
G2	85.5 $\pm$ 2.4	97.9 $\pm$ 0.2	14.5	1.06
G3	136.0 $\pm$ 3.9	149.0 $\pm$ 0.6	9.6	0.70
G4	103.4 $\pm$ 1.5	123.6 $\pm$ 0.6	9.0 ^a	0.66 ^a
G5	133.7 $\pm$ 2.0	153.6 $\pm$ 1.2	14.9	1.08

^a Repeated spawning of G2 Control parents corrected for the response in G3.

3.2. Realized genetic gain in BLUP breeding values

Average BLUP breeding values for body weight at harvest in grams of all individuals recorded in each generation of the selection line and in each of the control groups are shown in Fig. 2. The genetic trend in the selection line was linear, with an accumulated realized response in BLUP breeding values of 29.4 g over the five generations of selection. The average standard deviation of BLUP breeding values within selection line generations was 3.7 g. Consequently, the average realized response to selection per generation was 1.59 BLUP standard deviation units.

Since the animal effect in the BLUP model is a random effect, the breeding value solutions are expressed relative to a mean value of zero in the first generation included in the pedigree records (Gjerde, 2005), i.e. the parents of the G0 individuals. The mean body weight at harvest of sex by test environment sub-groups in this first generation (the diallel cross population) was 43.8 g (Bentsen et al., 1998). Considering the realized response in BLUP breeding values of 29.4 g, this suggests an accumulated genetic gain in BLUP breeding values relative to the mean body weight at harvest of the parent generation of G0 of 67% or somewhat lower than the estimate of 88% genetic gain based on control group comparisons (Section 3.1). Furthermore, since the parents of G0 were not chosen entirely at random (see Section 2.1 above), this estimate may include some response to selection within the parent generation of G0 that is not included in the estimate based on control group comparisons.

As expected, the mean BLUP breeding values of the control groups of progeny from matings between random G0 parents, or between genetically average G1, G2 and G4 individuals from the selection line (see Section 2.5 above), were very similar to the mean breeding values of their parents' generation, and so was the progeny of the re-mated average individuals one generation later (Fig. 2).

3.3. Ancestry from founder strains and rate of inbreeding

The proportion of purebred founder strain ancestors in each generation is shown in Table 6. In G0, the largest ancestry proportion was from the Kenya strain (27%), followed by the Egypt and Thailand strains (20% each) and the Senegal strain (12%). The remaining strains were all represented among the purebred founder ancestors, but in smaller proportions (3–9%). After five generations of selection, the ancestry proportion of the four most represented strains had increased from 79% to 91%, while the ancestry proportion of the remaining strains was reduced from 21% to 9%. However, all eight strains were still represented in G5 (Table 6).

The mean coefficient of inbreeding in each generation is shown in Table 6. As expected from the mating design of the base population parents (three- or four-way strain crosses), the coefficient of inbreeding was zero in all G0 individuals. The average increase of the coefficient of inbreeding was 1.43% per generation, corresponding to an average coefficient of inbreeding of 7.13% in G5. However, a significant proportion of the increase of inbreeding occurred during the mating of the parents of G2 (2.70%), while the average rate of inbreeding in the remaining generations was 1.11%.

4. Discussion

4.1. Response to selection

Prior to the initiation of the GIFT project, several attempts to improve growth rate in Nile tilapia by mass selection were reported unsuccessful (Huang and Liao, 1990; Hulata et al., 1986; Teichert-Coddington and Smitherman, 1988). A modified mass selection approach involving early size grading or culling of large fry (collimation) resulted in a modest response of 2–3% per generation (Basiao and Doyle, 1999; Brzeski and Doyle, 1995). On the other hand, the application of updated animal

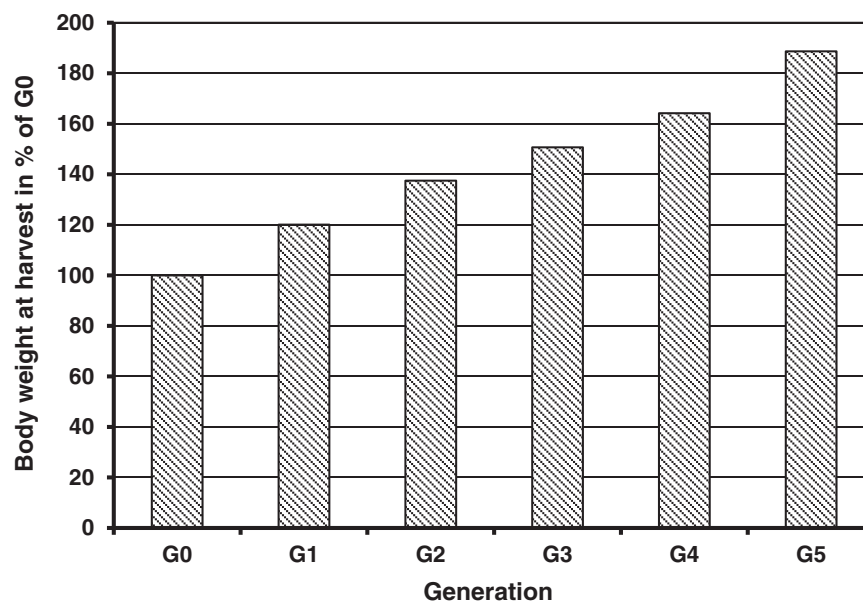


Fig. 1. Realized response to selection for increased body weight at harvest in percent of G0, estimated as the within generation difference between the least square means of the selection line and a control group of progeny of breeders with an expected average genetic merit in the previous generation.

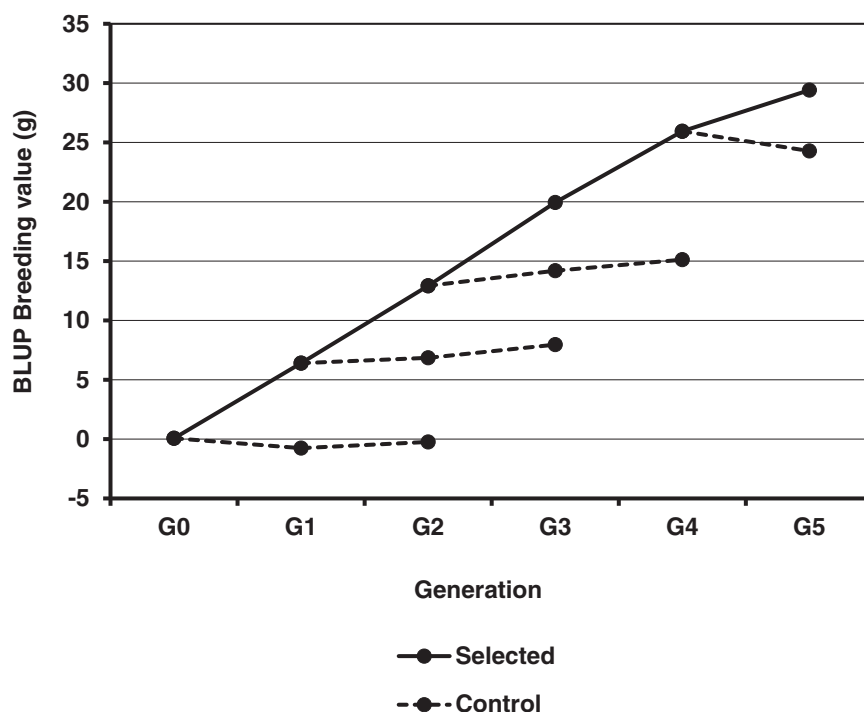


Fig. 2. Average BLUP breeding values for body weight at harvest computed across all generations of selection and all control groups. The control group in two following generations descending from the same selection line generation were both direct progeny of the same selection line breeders with an expected average genetic merit for body weight at harvest in the parent generation.

breeding technologies (estimation of breeding values based on records from individually tagged and pedigreed individuals) to farmed Atlantic salmon in Norway since the 1970-ies had resulted in an average increase of body weight at harvest of 14% per generation during several generations of selection (reviewed by Gjerdem, 2010). It has been argued that the poor response to mass selection in Nile tilapia may have been partly due to the lack of opportunity to correct for age-dependent body weight variation and other systematic sources of variation. It may also partly have been due to reproduction by mass-spawning prior to and during the experiments that may have resulted in highly unbalanced family structures and accumulation of inbreeding that would reduce the genetic variation of the populations and result in inbreeding depression. (Bentsen and Olesen, 2002).

Based on the experiences from the Norwegian salmon selection program, the GIFT project developed methods for single pair mating of Nile tilapia, separate rearing of full-sib progeny groups and individual tagging and pedigree recording (Eknath et al., 1993, 2007). These techniques made it possible to correct the body weight records for systematic environmental effects and thus separate these from additive genetic effects, and also to obtain breeding values with higher accuracy by including information of full- and halfsibs of the breeding candidates. The genetic variability of the base population was ensured by creating a composite population of the eight founder strains (Eknath et al., 2007). This resulted in an estimated average realized response to selection in

body weight at harvest of 13.6% or 0.99 additive genetic standard deviations per generation over five generations relative to the internal control groups, similar to the response magnitude obtained in the Norwegian salmon selection program. The BLUP breeding values suggested an accumulated selection response in body weight at harvest after five generations of selection of 67% relative to the parent generation of G0. This value was lower than the estimated realized accumulated genetic gain of 88% relative to G0 based on comparisons with control groups. The BLUP analysis confirmed that the control groups were representative of the average breeding values in the previous generation of the selection line (Fig. 2). However, the accuracy of the estimated breeding value of the G1 control group was probably poorer than that of the control groups in later generations. The genetic ties between this control group and the selection line was weaker; only related by sharing a low number of random parents with the G0 generation (11 sires and 12 dams); while the control groups in later generations were direct progeny of larger numbers of average breeders in previous selection line generations (Table 4). Still, even if the estimated realized response between G0 and G1 is omitted, the average realized response per generation between G1 and G5 was 12.0% (Table 5), and the accumulated realized response across the four generations relative to G1 was 64%. The use of progeny of parents with average breeding values in each generation as internal controls is expected to be more robust against accumulated genetic drift and unintended selection than a separate control line,

Table 6

The proportion of purebred founder strain ancestors of each generation, and the average coefficient of inbreeding (F) in each generation.

Generation	Proportion of purebred founder strain ancestors (%)								F (%)
	Egypt	Ghana	Kenya	Senegal	Israel	Singapore	Taiwan	Thailand	
G0	20.2	2.6	27.3	12.2	3.4	6.3	8.5	19.6	0.00
G1	25.7	0.4	27.1	17.7	4.8	1.5	6.6	16.2	0.15
G2	25.7	0.6	28.8	18.0	2.0	1.6	6.4	16.9	3.85
G3	25.8	0.4	31.3	19.7	1.6	0.8	4.9	15.5	5.34
G4	26.5	0.7	29.9	19.8	1.8	0.7	5.3	15.4	7.16
G5	26.4	0.6	30.1	19.8	1.9	0.7	5.3	15.2	7.13

and the method is cost-effective in terms of both facility and labor requirements (Rye and Gjedrem, 2005).

In 2005, after the termination of the GIFT project, cryopreserved spermatozoa from 13 males selected for high growth performance among the G0 test fish were used to produce progeny that were tested along with progeny of 13 average males from the ninth generation in the continued GIFT selection program carried out by WorldFish in Malaysia (Khaw et al., 2008). Considering that the G0 males were already selected, and that selection was suspended for two generations during the transfer of the GIFT material to Malaysia (Khaw et al., 2008; Ponzoni et al., 2005), the comparison covered six generations of selection. The estimated total genetic progress in body weight at harvest was 64%, which was lower than the estimated realized progress of 88% over the five generations in the present study. However, the low number of males used in the comparison study implies a risk of sampling bias. In particular, the selected G0 males may not have been representative for the average performance of all selected G0 breeders that contributed as parents to the G1 generation. Considering the various estimates, it may be concluded that the response to selection during the five generations in the GIFT project was in the range of 10% to 14% per generation.

After the transfer of the GIFT strain to Malaysia, selection was restarted in 2002 following a similar design to that used in the GIFT project. The accumulated response to selection relative to the 2002 population across the nine following generations of selection was 107% or 11.9% per generation (Hamzah et al., 2014), or slightly lower than in the initial five generations reported here. Approaches similar to the one used in the GIFT project have also since then been applied to several other GIFT-related populations (Gjerde et al., 2012; Hussain et al., 2011; Luan, 2010; Thodesen et al., 2011, 2012; Trong et al., 2013a) as well as other Nile tilapia populations (Charo-Karisa et al., 2006; Gall and Bakar, 2002; Rezk et al., 2009) and populations of other tilapia species (Maluwa and Gjerde, 2007; Thodesen et al., 2013a, 2013b; Zak et al., 2014). Average generation by generation response across the six of these studies that applied single trait selection for growth performance in Nile tilapia was 11.7% (15 generations, range 5.9–34.7%), which agrees well with the results in this study. Other approaches reported to result in significant response to selection are within family selection (Bolivar and Newkirk, 2002) and mass selection with early culling of large fry (Basiao et al., 2005). However, within family selection will not utilize the additive genetic variation that occurs between families to generate response to selection, and mass selection over several generations will require additional measures like rotational cohort mating (Ponzoni et al., 2010a) or single pair mating with separate hatching and standardization of family sizes (Bentsen and Olesen, 2002; Gjerde et al., 1996) to control the rate of inbreeding.

The estimates of the effects common to full-sibs other than additive genetic effects on body weight at harvest were relatively high in the present material, averaging 10% of the total variation in harvest body weight ($c^2 = 0.10$, Bentsen et al., 2012). This agrees well with estimates from the other previously referred studies of Nile tilapia. A major source of this effect is probably the long period of separate rearing of the full-sib groups until they reach a suitable tagging size of 3–5 g. In the present study, full-sib groups were reared separately until an average age of about 90 days, and body weight at harvest was then recorded after communal grow-out until an average age of about 200 days (Table 1). Even though efforts were made to standardize the treatment and environment during the separate rearing of the full-sib groups, random differences were inevitable. The impact of such common effects within full-sib groups on harvest body weight will be confounded with the additive genetic effects, in particular when selection is based on breeding values computed from information from contemporary relatives only, as in the present study, resulting in lower heritability and breeding values with low accuracy. The genetic gain may consequently be increased if the c^2 effects could be reduced. A modification of the GIFT design was introduced by Thodesen et al. (2011), where full-sib families were reared

and tested in several separate batches depending on the timing of spawning in the parent broodstock. This reduced the period of separate rearing to about 60 days, but did not seem to reduce the c^2 effects. Trong et al. (2013a) used a so-called 'single male-multiple females' mating approach that reduced the time to produce the necessary half-sib groups to 30–45 days. However, as the costs and speed of DNA-based ancestry determination techniques are improved (Fu et al., 2013; Trong et al., 2013b), communal stocking of full-sib families from the egg stage or as swim-up fry would probably reduce the c^2 effects significantly, as demonstrated for growth in common carp (Ninh et al., 2011) with a substantial increase in the response to selection (Ninh et al., 2013).

4.2. Effects of selection on the pure strain ancestry of the population

The aim of the mating design of the parents of the base population (three- and four-way strain crosses between parents from the best growing diallel strain combinations and representation of all strains, Eknath et al., 2007) was to establish a fast growing base population with genetic contributions from all eight founder strains. It was known from the pure strain test that the growth performance of founder strains differed considerably, and the geographic origin (Eknath et al., 1993) and the genetic relationship (Macaranas et al., 1995) of the strains were highly diverse. It was considered likely that the strains might differ in traits other than growth performance. Such traits could become important goals in future selection programs (e.g. survival, disease resistance, sexual maturation, carcass quality), and some of the slow growing strains could possess valuable genetic variation for such traits.

When the Norwegian Salmon selection program was initiated in the early 1970s, 41 different founder populations were tested. Selection for improved growth performance was then carried out across populations without restrictions regarding the strain origin of the breeding candidates. This resulted in the elimination of the contributions from most of the original sub-populations during the first couple of generations of selection when the new farmed populations were formed (Gjedrem et al., 1991).

In the present study, two generations of systematic strain crossing and relaxed selection resulted in a base population of highly crossbred individuals. The representation of ancestors from each of the eight original founder strains may be used as an estimate of the genetic composition of each generation. It should be noted that the ancestor representation does not necessarily give an exact measure of the genetic composition of the selection line generations, since the realized ancestry is affected by random recombination and selection (i.e. contemporary descendants of the same founder strain combinations may vary in their real genetic strain composition). The proportion of ancestors from each of the founder strains when traced back from the base population individuals (G0, Table 6) was higher for the fast growing strains than the slower growing strains (Eknath et al., 1993). Still, the ancestor representation from the slower growing strains was further reduced during the first couple of generations of selection until an apparent steady representation was reached (Table 6). The ancestor representation from three of the founder strains (Ghana, Israel and Singapore) was then marginal (0.6–1.9% in G5). However, very few of the parents of G5 had more than two ancestors from each of these founder strains. Consequently, the ancestor representation from each strain was distributed over 25% or more of the G5 parents. This made the ancestry contributions robust against random extinction, and probably still sufficient to respond to changes in the artificial or natural selection pressure. In general, significant additive genetic variation was found for a range of traits other than harvest body weight, both in the presently reported GIFT population (e.g. for frequency of early sexually maturing females, Longalong et al., 1999; for serum lysozyme activity, Chiayvareesajja et al., 1999; for male proportion among progeny, Lozano et al., 2013), and in descending populations (e.g. for fillet yield, Gjerde et al., 2012; Nguyen et al., 2010; Thodesen et al., 2012, for body shape, Trong et al.,

2013c, for survival in freshwater and brackish water and for cold tolerance, Luan, 2010, for female reproductive traits, Trøng et al., 2013a, 2013d).

Note that the most successful founder strains measured by their increased ancestor representation during the five generations of selection, were all wild African strains (the Kenya, Egypt and Senegal strains, from 59.7% representation in G0 to 76.3% in G5), while the farmed Asian founder strains were all less successful. This may reflect the history of the introduction of Nile tilapia in Asian aquaculture during the 1980s. The farmed strains were descendants of a few introductions of small numbers of fish from Africa, mostly through intermediate non-tropical countries, and were likely to be suffering from genetic founder and bottleneck effects (Pullin and Capili, 1988), genetic deterioration due to introgression of genes from *Oreochromis mossambicus* (Macaranas et al., 1986; Macaranas et al., 1995). In the growth performance test of the original pure founder strains, the Kenya and Egypt strains had the highest body weights at harvest, followed by the Thailand and Senegal strains. The poorest performers were the Israel and Ghana strains (Eknath et al., 1993).

4.3. Inbreeding

The increase of the coefficient of inbreeding per generation in the present study (1.43%) was within the range of estimates from the previously cited reports on selection in Nile tilapia (0.5–2.0%). Estimates of phenotypic coefficients of variation and heritabilities for body weight at harvest (Bentsen et al., 2012) suggested that the additive genetic variation was sustained during selection. In several descending populations, response to selection has continued during up to ten generations or more (Hamzah et al., 2014; Jantøy, 2013; Thodesen et al., 2011;).

However, inbreeding was slightly higher than recommended in long term selection programs for terrestrial livestock species (0.5–1.0% per generation, Bijma and Woolliams, 2000). The long term coefficient of inbreeding may be reduced by increasing the effective population size, i.e. by further restriction on the maximum number of breeders selected from each full-sib group (in the present study; less than the applied limit of maximum 15 selected full-sibs, and/or imposing an additional restriction on the maximum number selected from each half-sib group. The nested mating design used in the present study may also require specific limitations on the number of males used from each full-sib group). Furthermore, a significant part of the increase of inbreeding in the present study happened during one generation (2.70% between G1 and G2), suggesting that the restriction method applied during mating in the present study (no mating between full- or half-sibs) may not be sufficient to prevent episodes of increased inbreeding. Ponzoni et al. (2010b) applied a mate allocation strategy where the inbreeding coefficient of potential progeny of tentative matings were checked and only matings resulting in inbreeding coefficients below a fixed limit were accepted. The limit was set to 0% in early generations and 3% in later generations. As far as possible, the contribution from each family was restricted to one male and two females. The rate of inbreeding was 0.0037 per generation and it was concluded that the mate allocation strategy was successful in keeping the rate of inbreeding at an acceptable level. A similar strategy was used in the 5–6 initial generations of the selection programs for Atlantic salmon and rainbow trout in Norway (Gjerde, pers. comm). The application of more advanced approaches like optimum contribution selection methods may offer a more rigorous control of inbreeding (Meuwissen, 1997; Sonesson et al., 2005). Furthermore, as the number of genetically divergent, improved populations of Nile tilapia increases, accumulation of inbreeding may be counteracted by interbreeding between populations.

4.4. Dissemination of genetic progress

The GIFT project was a global UNDP project intended to provide improved genetic material and technology to fish farmers worldwide,

including small scale and backyard farmers, with The WorldFish Center (then ICLARM) as the executing agency. The choice of a purebreeding selection program for long term improvement of additive genetic performance within a single heterogeneous synthetic population, rather than a hybrid program involving selection within and crossing between several separate strains, was partly based on the relatively insignificant and environmentally sensitive heterosis effects observed in the initial diallel cross experiment (Bentsen et al., 1998). Furthermore, it is also much easier to disseminate additive genetic gain from a purebreeding selection program to the target farmers than to also disseminate heterosis in a hybrid program. Broodstock from the selection program nucleus may be used directly by multiplier operations to produce grow-out fry without complicated hybrid mating schemes, as long as inbreeding is restricted by supplying replacement broodstock from a wide range of families from each new generation in the breeding nucleus. Because of the high fecundity of farmed fish compared to terrestrial farm animals, the genetic progress in the breeding nucleus may then reach the grow-out farmers with a delay of only one generation. The full benefit of this will only be achieved if both multipliers and grow-out farmers replace their stock as soon as a new, improved generation is available. However, farmers that depend on on-farm reproduction for economic or logistic reasons may also benefit from the genetic progress in a pure selection program, as long as they replace their broodstock with individuals from improved generations at regular intervals. Between replacements, appropriate on-farm reproduction to avoid inbreeding is expected to maintain the additive genetic performance of the latest replacement. A hybrid program will be less beneficial for this type of farmers, since on-farm mating of hybrids is expected to result in recombination loss of performance.

Genetic material from the GIFT project was disseminated through ICLARM for on-farm testing from the very early generations in 1994, both within the Philippines and through public institutions in Bangladesh, China, Thailand and Vietnam (Dey et al., 2000). However, since the development of high performing breeds of farm animals is a long-term process of continuous additive genetic improvement through repeated selection (Bentsen, 1990; Eknath et al., 1991; Gjedrem et al., 2012; Ponzoni et al., 2011), farmers will depend on continued access to updated generations from long-term selection programs to gradually improve their production. The ultimate benefit of a project like GIFT will consequently be the establishment of viable and continuous programs of selective breeding and dissemination. During the last couple of years of the GIFT project, private Philippine operators were trained and accredited to multiply and market GIFT seed, which became an immediate success among tilapia farmers in the Philippines (Yosef, 2009). In 1997, a random sample of on average 200 full-sibs from each of 106 G5 full-sib families were supplied to Research Institute for Aquaculture No. 1 (RIA1) Hanoi, Vietnam to found a further selection program there (Luan, 2010; Ponzoni et al., 2010a).

When the external project funding ended in 1998, a non-profit foundation (The GIFT Foundation International, GFII) was formed at the project site in Muñoz to secure the GIFT population (including selected broodstock from the G5 generation in the present study, i.e. G6 parents) and to continue the selection program. However, the revenues generated by the sales of the improved stock remained largely with the multipliers and farmers, and failed to flow back to the breeding nucleus (Acosta et al., 2006; Eknath and Hulata, 2009; Ponzoni et al., 2010a). The problems of benefit sharing along the value chain of aquaculture selection programs are well known (Olesen et al., 2007). In the absence of further public funding, GFII faced shut-down. An agreement was then made with a private Norwegian company (GenoMar, then BioSoft) to fund the basic operation of GFII. In return, GenoMar was granted access in 1999 to pedigreed, unselected second generation progeny of the selected G5 broodstock (Eknath and Hulata, 2009; Ponzoni et al., 2010a), or G6 in the terms of the present study (counting generations of selection). Counting all generations, including the initial pre-selection GIFT generations, this material was also referred to as G10.

Furthermore, a pedigreed set of G6 full-sib families from 63 single pair matings (1:1) of selected G5 parents was transferred from the GFII facilities to WorldFish in Malaysia in 2000–2001 (Ponzoni et al., 2005, 2010a). After the termination of the GIFT project, GFII has focused mainly on production of broodstock and seed for the Philippine tilapia industry.

A new G6 GIFT base population was formed in 2002 by WorldFish in Malaysia by mating broodstock reared from the fish transferred from GFII, and a further pedigree based selection program for improved growth performance with a one-year generation interval was initiated in collaboration with Department of Fisheries, Malaysia at the Aquaculture Extension Center in Jitra, Kedah State (Ponzoni et al., 2005). In addition to supplying tilapia farmers in Malaysia with continuously upgraded seed, WorldFish is also collaborating with public and semi-public institutions in other countries to transfer the most recent GIFT material and to initiate further local selection and/or dissemination programs in the recipient institutions. Both pedigree based selection and modified mass selection with rotational cohort mating have been applied (Ponzoni et al., 2010a). So far, this has involved transfers to Bangladesh (Bangladesh Fisheries Research Institute, Mymensingh, 1994–2005, BRAC Center, Magura, 2008), Brazil (Universidade Estadual de Maringá, 2005), China (Shanghai Ocean University, 1994, Freshwater Fisheries Research Center, Wuxi, 2006), Philippines (TGA Farms Incorporated, Pampanga, 2006), Sri Lanka (Aquaculture Development Center, Dambulla, 2007), Thailand (Pathumthani Fisheries Test and Research Center, 1994–2007) and Vietnam (RIA1, Hanoi, see above, RIA2, Ho Chi Minh City, 2006) (Ponzoni et al., 2010a). The creation of a network of GIFT breeding populations to increase the effective population size by exchange of broodstock, similar to breed associations of terrestrial farm animal species, has been suggested (Ponzoni et al., 2010b).

Genetic material and technology from the GIFT project has also been a source for private breeding operations. GenoMar used the material acquired in 1999 to establish a G6 breeding nucleus in Luzon, Philippines and has since then carried out a pedigree based selection program with a one-year generation interval for improved growth rate and fillet yield. The seed is marketed under the brand GenoMar Supreme Tilapia (GST) in the Philippines and through multiplier operations in Hainan, China and Lake Temenggor, Malaysia (Jamtøy, 2013). RIA1 in Vietnam used the material acquired in 1997 to establish a national mixed population initially based on 80% G5 GIFT broodstock and 20% local broodstock (Ponzoni et al., 2010a) under the brand NOVIT, that has been pedigree selected since then for growth rate and cold tolerance and marketed in Vietnam and abroad. In 2004, 100 pedigreed NOVIT full-sib families from the fifth generation of selection (i.e. G10 in a GIFT context) was exported to China and used there to establish a pedigree based selection and multiplier program with one-year generation intervals for improved growth rate and fillet yield in Hainan, China under the brand Progift (Thodesen et al., 2011, 2012). The same year, 100 pedigreed NOVIT families from the fourth generation of selection were also exported to Managua, Nicaragua, and served as base for three generations of selection in a similarly structured in-house selection program operated by the company Nicanor S.A. (Gjerde et al., 2012). Material from the Nicaraguan program was then transferred to Florida, USA, in 2010, and used to establish a pedigree based selection and multiplier program under the brand Spring Genetics. Spring Genetics seed is currently marketed in USA and worldwide, and through multiplier operations in Mazatlán, Mexico and Fortaleza, Brazil (Spring Genetics, 2016). The mentioned pedigreed selection programs in China, Nicaragua (now terminated) and USA were all designed and supervised by Akvaforsk Genetics Center (AFGC), a Norwegian based provider of genetic improvement services to aquaculture industries. Furthermore, since GIFT-descending genetic material has been available on the open market during many years, it is not unlikely that unrecorded use of the material and technology has contributed to the establishment of other commercial breeding populations as well.

So far, the GIFT project has had limited impact in Africa. From the early years of the project, there was international concern about possible environmental impacts of a systematically selected, genetically mixed tilapia population for general farming purposes (e.g. Doyle et al., 1991). One of the concerns was associated to re-introductions of GIFT-tilapia for farming in Africa and possible interbreeding between GIFT-escapees and wild tilapia resource populations. Consequently, caution was recommended regarding re-introductions of GIFT genetic material to Africa (Bentsen et al., 1992; Pullin, 1994). As an alternative, it was recommended to prioritize the transfer of GIFT technology to develop farmed populations in Africa based on local stocks. Several projects based on the GIFT approach have been initiated by WorldFish in Africa since then, e.g. in Egypt (Rezk et al., 2009), in Ghana (Lind et al., 2012; Ponzoni et al., 2010a; Tran et al., 2012), and in Malawi for *Oreochromis shiranus* (Maluwa and Gjerde, 2007). Bearing in mind that ten years of extensive external funding was not sufficient to secure the viability of the GIFT project, it is evident that the development of an aquaculture industry in Africa entirely based on local selection programs will be an expensive and time consuming strategy. Meanwhile, selection programs outside Africa have accumulated genetic progress and competitive advantage during > 15 generations of selection, encouraging further unauthorized introductions to Africa. Since 2007, it has been recommended by WorldFish to support authorized, risk assessed and monitored re-introductions of GIFT material to Africa (WorldFish, 2007). The first case study is now being implemented in Ghana (Rosendal et al., 2012; Tran et al., 2012). Preliminary results from a comparison between GIFT and the locally improved Akosombo strain of Nile tilapia in Ghana indicate an advantage of GIFT of the order of 100% in harvest weight, and no difference between the strains in survival rate (unpublished results). The difference in harvest weight is large enough to justify the positive impact that rigorously controlled dissemination of the GIFT strain in Africa could have.

5. Conclusions

The GIFT project showed that the application of up to date methods of traditional farm animal selection technology to a genetically variable population of Nile tilapia resulted in considerable genetic response in growth rate (in the range of 10% to 14% per generation) across a wide range of farm environments during five generations of selection. However, the project also confirmed that the development of aquaculture selection programs requires long term external funding. Despite considerable economic benefits on a society level, it has been difficult to secure the commercial viability of the GIFT program based on revenues from sales of improved broodstock only. Still, descending populations from the original GIFT material have since the end of the project been further selected and have performed well under a wide range of farming conditions worldwide, and the technology has been successfully applied by public as well as private breeding operations.

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